Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/17/2016
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Cognitive Performance and Crew Cohesion during Confinement in NASA's Human Research Program Human Exploration Research Analog (HERA)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/24/2014	**End Date:**	08/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per NSSC information (Ed., 5/16/17) NOTE: Extended to 5/31/2017 per PI (Ed., 2/3/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17) NOTE: Extended to 3/31/2017 per NSSC information (Ed., 10/5/16) NOTE: Extended to 9/30/2016 per information from D. Arias/HRP BHP (Ed., 2/17/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gur, Ruben Ph.D. (University of Pennsylvania, School of Medicine) Dinges, David Ph.D. (University of Pennsylvania, School of Medicine)		
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Performance Goal Text:			

Task Description:	The recently established Human Exploration Research Analog (HERA) plans for campaigns of incremental duration starting early 2014. A campaign is defined as one integrated protocol with one primary mission scenario. A campaign will consist of multiple missions in order to meet study subject requirements. Mission durations are planned from 7 days up to 30 days. The HERA planning schedule currently anticipates 4 missions per year (one per quarter), of 7-day duration in 2014, 14-day duration in 2015, and 28-day duration in 2016. This project seeks to be implemented in 4 missions of campaign 1 (7-day missions) and in 4 missions of campaign 2 (14-day missions). Human exploration of space requires astronauts to maintain a high level of cognitive performance in the presence of a range of fatigue-related conditions associated with perturbations of sleep and circadian neurobiology (e.g., slam shifts), excessive cognitive and physical workloads (e.g., EVAs), and environmental stressors (e.g., elevated CO2, hypoxia, radiation). This highlights the critical need for comprehensive, sensitive, and valid neurocognitive assessments in spaceflight in general and during exploration type missions specifically to guarantee mission success. This project will systematically assess cognitive performance in HERA crewmembers that are confined to the HERA facility for 7-14 days. HERA crewmembers will be exposed to different stressors including high and low workload, sleep deprivation, and time delays. The Cognition test battery consists of 10 validated and brief neuropsychological tests that cover a wide array of cognitive domains and that were specifically developed for high-performing astronauts by the research team. Our research in HERA has the following specific aims: Specific Aim 1: Test feasibility of regular Cognition data acquisition in the space analog environment of HERA. The Cognition software will be installed on Apple iPads. Before pre-isolation baseline testing, the crew will be asked to watch a standardized introductory video that familiarizes them with the Cognition software. They will then perform the battery (i.e., all 10 tests) once for familiarization, and will be asked to perform all practice tests during this first administration. The crew of 4 will then perform Cognition 3 times pre-mission, on a daily basis in-mission (in the 2-hour period pre bed), and 3 times post-mission. The Principal Investigator (PI) and his team will download the data and perform quality control checks shortly after the completion of each Cognition battery. This way, both technical and motivational problems can be addressed immediately. The crew will be de-briefed by the PI and his team by teleconference after the last post-mission test administration. Specific Aim 2: Perform descriptive analysis of changes in cognitive performance with time in mission and related to mission stressors. Key variables of the 10 cognitive performance tests will be analyzed descriptively and plotted by time in mission for each of the 4 HERA missions. Test raw data will be provided to NASA's Life Sciences Data Archive, and thus shared with other investigators. This way, individual investigators can analyze cognitive performance data related to mission events and interventions of specific experiments. Specific Aim 3: Measure rest activity patterns and light exposure with actigraphs (campaign 1 only). The crewmembers will continuously wear wrist actigraphs to determine rest-activity patterns and light intensity throughout the missions. These data will be used to characterize average sleep duration, the placement of sleep periods (naps?), the overall level of activity, and the intensity of light exposure depending on time in mission and mission specific events. Specific Aim 4: Test feasibility of proximity data acquisition in HERA (campaign 1 only). Seven beacons will be strategically placed throughout the facility (3 on floor 1, 3 on floor 2, and 1 in the hygiene module). Each crewmember will wear one beacon and one receiver continuously throughout the mission. We will investigate whether we will be able to detect where in the facility and with whom crewmembers spend most of their time. As this technology is very new, this specific aim is exploratory. Project Methodology. Subjects and Design. For each of the 4 one-week missions, 4 mixed-gender crews will be selected by NASA. The crewmembers will be NASA employees that serve as astronaut surrogates and pass the applicable health screenings. Both NASA and UPenn IRB (Institutional Review Board) approval have been obtained. Cognition test battery: The Cognition test battery consists of 10 brief neuropsychological tests that cover a wide range of cognitive domains (executive control, memory, attention, emotional processing, risk decision making, abstraction, sensorimotor speed) that include and go beyond what is currently measured with NASA's WinSCAT battery. Importantly, brain regions involved in performing each of the Cognition tests have already been established with fMRI, and the tests that are the basis for Cognition have been well-validated in both healthy individuals and patient populations. Cognition can be performed on PCs/laptops running Windows 7 or higher, or on iPads/iPad minis. This assures maximal flexibility in test administration, and no up mass for spaceflight.
Rationale for HRP Directed Research:	Dr. Basner and his team have developed the Cognition battery specifically for NASA and the astronaut population. Ground and space flight testing of the battery are in the process of data collection. To consider transitioning such a tool to an operational use, further validation and testing is required. This would be a requirement of any new cognitive battery. The highly specialized testing that is offered by the proposed recipient is unique and to date we have not found a cognitive toolkit that provides the level of specificity and usability as the Cognition toolkit. Since all the work has been done my Dr. Basner to date, it is not feasible to begin with a new contractor. The costs to start over would likely match those spent with University of Pennsylvania to date. This is a very low cost effort. The majority of the budget is directed towards labor/personnel which has been deemed reasonable for the amount of hours proposed.
Research Impact/Earth Benefits:	The Cognition test battery was specifically designed for the high performing and highly motivated astronaut population. It will thus be a useful tool for the objective assessment of cognitive performance in similar high performing populations on Earth (e.g., medical doctors).
Task Progress:	Cognition data acquisition in all 16 crewmembers of campaign 1 and all 16 crewmembers of campaign 2 is complete. Actigraphy and proximity data have been successfully acquired in all 16 crewmembers during campaign 1. The data are currently being analyzed.

Bibliography Type:	Description: (Last Updated: 06/17/2026)
Abstracts for Journals and Proceedings	Nasrini J, Dinges DF, McGuire S, Hermosillo E, Ecker AJ, Mollicone DJ, Mott CG, Port A, Moore TM, Gur RC, Basner M. "Cognition in a Human Exploration Research Analog (HERA)." Poster at 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Articles in Peer-reviewed Journals	Basner M, Savitt A, Moore TM, Port AM, McGuire S, Ecker AJ, Nasrini J, Mollicone J, Mott CM, McCann T, Dinges DF, Gur RC. "Development and validation of the Cognition test battery for spaceflight." Aerospace Medicine and Human Performance. 2015 Nov;86(11):942-52. http://dx.doi.org/10.3357/AMHP.4343.2015 ; PubMed PMID: 26564759; PubMed Central PMCID: PMC4691281 , Nov-2015